

EarthMind-4B

Baseline Evaluation Report - SatQuery AI

Prepared by	Karan Dalal - SatQuery AI Team
Model	EarthMind-4B (sy1998/EarthMind-4B)
HuggingFace	huggingface.co/sy1998/EarthMind-4B
Paper	arxiv.org/abs/2506.01667
Date	September 2026
Recommendation	GO - Primary model for SatQuery AI

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Section 1: Model Overview - EarthMind-4B

EarthMind-4B is a multi-modal, multi-sensor Vision-Language Model (VLM) developed specifically for Earth Observation (EO) tasks. Unlike general-purpose VLMs that are trained on internet photos, EarthMind was pre-trained on Sentinel-1 SAR radar imagery and Sentinel-2 multispectral optical imagery - the same sensor types used in real-world satellite analysis.

1.1 Model Details

Property	Value
Model Name	EarthMind-4B
Parameters	4 Billion
HuggingFace Link	huggingface.co/sy1998/EarthMind-4B
Research Paper	arxiv.org/abs/2506.01667 - EarthMind: Multi-modal Multi-sensor Earth Observation VLM
Training Data	Sentinel-1 (SAR/Radar) + Sentinel-2 (Optical/Multispectral) satellite imagery
Quantization	4-bit NF4 via BitsAndBytesConfig - fits in RTX 4060 8GB VRAM
Inference VRAM	Approximately 3-4GB with 4-bit quantization

1.2 What Makes EarthMind Different

Most VLMs (GPT-4V, LLaVA, InternVL3) are trained on natural photographs from the internet. Satellite images are fundamentally different - they are captured from hundreds of kilometers above, use multiple spectral bands beyond visible light, and require understanding of land cover patterns, geographic features, and temporal changes. EarthMind was designed from the ground up to handle these satellite-specific characteristics.

Feature	General VLMs	EarthMind-4B
Training images	Internet photos	Sentinel-1 and Sentinel-2 satellite data
SAR support	No	Yes - handles radar imagery natively
Sensor fusion	Single modality only	Optical + SAR joint analysis
Change detection	Limited	Bi-temporal image pair reasoning
Domain vocab	General English	LULC, spectral bands, satellite terms
VQA accuracy	~60% on satellite data	72% without any fine-tuning

Section 2: Why EarthMind-4B Was Selected

Reason	Explanation
Satellite Native	Pre-trained on Sentinel-1 + Sentinel-2. No domain gap between training and deployment data.
SAR + Optical	Handles both radar (SAR) and optical images. Required by SIH PS for cross-modal analysis.
97.2% VQA on AID	State-of-the-art benchmark results on aerial image datasets before any fine-tuning.
BigEarthNet Aligned	The EarthMind paper uses BigEarthNet.txt for evaluation - same dataset we use for benchmarking.
4B Parameters	Large enough for strong performance, small enough to run on RTX 4060 with 4-bit quantization.
Multi-task	Single model handles VQA, captioning, grounding, change detection, and SAR+optical fusion.
Open Source	Freely available on HuggingFace. No API costs for SIH demonstration.

Section 3: Datasets Used for Evaluation

Four datasets were used to evaluate EarthMind-4B across all five satellite image analysis tasks. Each dataset was downloaded from HuggingFace and stored locally. The test splits were used for evaluation only - no training was performed on EarthMind.

Dataset	HuggingFace	Task	Samples
VRSBench	xiang709/VRSBench	VQA + Captioning + Grounding	200 each task
RSVQA-HR-2k	dmarsili/RSVQA-HR-2k	VQA - Count, Area, Presence, Comparison	500
SECOND	EVER-Z/torchchange_second	Change VQA (bi-temporal pairs)	200 pairs
EarthMind-Benchmark	sy1998/EarthMind-Benchmark	Multi-task official benchmark	200

3.1 RSVQA Question Type Breakdown

RSVQA-HR-2k contains four types of questions. Breaking results by question type reveals specific weaknesses that cannot be seen from overall accuracy alone.

Type	Count	Answer Pattern	Example
Comparison	162	Yes or No	Are there more roads than buildings?
Presence	147	Yes or No	Is there a water area in the image?
Count	128	Whole number (e.g. 3, 7)	How many buildings are visible?
Area	63	Number + m2 (e.g. 495m2)	What area do the roads cover?
Total	500	-	-

Section 4: Evaluation Metrics Explained

4.1 VQA Metrics

Metric	Formula	Meaning
Exact Match	$\text{Correct} / \text{Total} \times 100$	Prediction must exactly match ground truth after lowercasing and removing punctuation.
Soft Accuracy	GT tokens found in pred / GT tokens	Handles verbose answers. "Yes there is a river" matches GT "yes" with soft match.
Numeric Accuracy	Matching numbers / Total	Extracts numbers from answers. Used for count (3 buildings) and area (495m2) questions.
SBERT Cosine	$\text{cosine}(\text{embed}(\text{pred}), \text{embed}(\text{gt}))$	Sentence-BERT semantic similarity. High score means same meaning even with different words.

4.2 Captioning Metrics

Metric	Formula	Meaning
BLEU-4	$\text{BP} \times \exp(0.25 \times \sum \log \text{n-gram precisions})$	Counts matching 1,2,3,4 word sequences. Strict - different wording gives low score.
ROUGE-L	Longest Common Subsequence / ref length	Measures sentence structure similarity. Less strict than BLEU on word order.
METEOR	F-score with exact + stem + synonym match	"Urban fabric" and "city area" get partial credit because they are synonyms.
CIDEr	TF-IDF weighted n-gram consensus	Rare satellite words like wetlands and arable get more weight than common words.
SBERT Cosine	$\text{cosine}(\text{embed}(\text{caption}), \text{embed}(\text{ref}))$	Semantic similarity of full caption. High means model described same things differently.

4.3 Grounding Metric

Metric	Formula	Meaning
mIoU	$\text{Mean}(\text{Intersection} / \text{Union}) \text{ across samples}$	How well predicted bounding box overlaps with ground truth. 0=no overlap, 1=perfect.
Accuracy@50	% samples with $\text{IoU} \geq 0.50$	Standard threshold. At least half of predicted box must overlap with ground truth.
Accuracy@25	% samples with $\text{IoU} \geq 0.25$	Lenient threshold. Even rough localization counts.
Parse Success	% samples with valid bbox format	Did model output $[x1,y1,x2,y2]$ correctly? 100% means always outputs correct format.

Section 5: Complete Benchmark Results

5.1 VQA Results

Dataset	Metric	Score	Status
VRSBench	Soft Accuracy	72.0%	Strong - above 60% GO threshold
VRSBench	Exact Accuracy	70.0%	Strong
VRSBench	SBERT Cosine	0.880	Very high semantic similarity
RSVQA-HR	Soft Accuracy	48.8%	Moderate - fine-tuning recommended
RSVQA-HR	Exact Accuracy	48.8%	Moderate
RSVQA-HR	SBERT Cosine	0.719	Moderate semantic similarity
RSVQA-HR	Numeric Accuracy	1.2%	Very weak - area and count tasks fail

5.2 Captioning Results

Metric	Score	Interpretation
BLEU-4	0.0845	Low exact n-gram overlap. Model uses different wording than reference captions.
ROUGE-L	0.2885	Moderate sequence similarity. Some sentence structure matches reference.
METEOR	0.3123	Good synonym-aware match. Model describes correct content with different words.
CIDEr	0.2591	Moderate domain keyword consensus. Satellite vocabulary partially present.
SBERT Cosine	0.7318	High semantic similarity. Model understands scene content well.

5.3 Grounding Results

Metric	Score	Interpretation
mIoU	0.3244	Moderate overlap. Model localizes regions with reasonable accuracy.
Accuracy@50	41.0%	41 of 100 predictions have more than 50% overlap with ground truth.
Accuracy@25	45.5%	45.5 of 100 predictions have more than 25% overlap with ground truth.
Parse Success	51.5%	Model outputs valid [x1,y1,x2,y2] format in 51.5% of cases. Room for improvement.

5.4 Change VQA Results

Metric	Score	Interpretation
Soft Accuracy	99.5%	Near perfect. Model correctly identifies change or no-change in bi-temporal pairs.
Exact Accuracy	99.5%	Matches ground truth exactly. Model gives concise correct answers.

5.5 Master Results Summary

Task	Dataset	Primary Metric	Score	Status
VQA	VRSBench	Soft Accuracy	72.0%	Strong
VQA	RSVQA-HR	Soft Accuracy	48.8%	Moderate
Captioning	VRSBench	BLEU-4	0.0845	Low - FT needed
Grounding	VRSBench	mIoU	0.3244	Moderate
Change VQA	SECOND	Soft Accuracy	99.5%	Excellent
Overall	-	Recommendation	GO	Primary Model

Section 6: Result Analysis and Key Findings

#	Finding
1	VRSBench VQA at 72% is strong without any fine-tuning, confirming EarthMind is the right choice as primary model. This exceeds the 60% GO threshold comfortably.
2	BLEU-4 (0.085) is low but METEOR (0.312) and SBERT (0.732) are high for captioning. This means the model describes correct satellite content but uses different vocabulary than reference captions. Fine-tuning on satellite-specific captions will improve BLEU.
3	Grounding parse success is 51.5% meaning the model fails to output a valid bounding box format in half the cases. When it does output a box, accuracy is reasonable (mIoU 0.32). Dhruv improved this using the SEG mask approach.
4	Change VQA at 99.5% is excellent. EarthMind understands bi-temporal satellite pairs very well because it was trained on Sentinel time-series data.
5	RSVQA Numeric accuracy is 1.2% - almost zero. The model cannot reliably extract numbers for count and area questions. This requires specific fine-tuning with numeric answer examples.
6	SBERT scores are consistently high across all tasks (0.72-0.88). This confirms EarthMind understands the semantic meaning of satellite scenes even when exact string matching metrics are lower.

Section 7: Failure Patterns and Fine-tuning Recommendations

7.1 Failure Patterns

Pattern	Score	Root Cause	Priority
RSVQA below 60%	48.8%	High resolution aerial images different from Sentinel training data	High
Captioning BLEU low	0.085	Model uses different vocabulary than reference captions. Not style-matched.	High
Grounding parse 51.5%	51.5%	Model inconsistently outputs bbox format. Prompt engineering helped partially.	Medium
Numeric accuracy 1.2%	1.2%	Model cannot estimate area in m2 or count objects precisely.	Medium
RSVQA Area 0%	0.0%	Complete failure on area estimation. No training data for this specific task.	High

7.2 Fine-tuning Recommendations

Priority	Task	Current	Target	Training Data
1 - Highest	Captioning	BLEU 0.085	BLEU 0.30+	BigEarthNet.txt - 9.6M satellite captions
2 - High	RSVQA VQA	48.8%	70%+	RSVQA-HR train split - focus on area and count types
3 - High	Numeric QA	1.2%	15%+	RSVQA with explicit number extraction examples
4 - Medium	Grounding	mIoU 0.32	mIoU 0.50+	VRSBench grounding train split + DIOR-RSVG

Section 8: Final Role in SatQuery AI

Based on all benchmark results, EarthMind-4B is confirmed as the primary model in the SatQuery AI system. It handles the majority of tasks with strong performance and is the only model in the pipeline with native SAR + Optical fusion support.

Task	Model	Score	Reason
VQA	EarthMind-4B	72.0%	Strong baseline. Highest VQA accuracy among all tested models.
Captioning	EarthMind-4B	BLEU 0.085	Best captioning metrics. SBERT 0.73 shows good semantic understanding.
Grounding	EarthMind-4B	mIoU 0.32	Best grounding accuracy. Acc@50 of 41% is 6x better than InternVL3 FT.
Change VQA	InternVL3 FT	100%	InternVL3 fine-tuned marginally better here. Lighter model for this task.
SAR + Optical	EarthMind-4B	Native	Only model with built-in SAR support. Required by SIH PS mandatory requirement.

8.1 How EarthMind Integrates with SatQuery AI

EarthMind-4B is loaded on demand by the `earthmind_runner.py` module using 4-bit quantization. The LangGraph agentic controller routes queries to EarthMind for VQA, captioning, grounding, and SAR+Optical tasks. For Change VQA, queries are routed to InternVL3-1B5 fine-tuned which is a lighter specialist model.

8.2 VRAM Management

On RTX 4060 8GB, EarthMind-4B uses approximately 3-4GB VRAM with 4-bit quantization. The model is loaded once at API startup and stays in memory for all subsequent requests. When switching to InternVL3 for Change VQA, EarthMind is unloaded first to free VRAM, then InternVL3 is loaded. This on-demand loading strategy keeps total VRAM usage within the 8GB limit at all times.